



Protocol on Data Exchange between Cash-Register and EFT-POS terminal

VERIFONE VX510, VX610, VX670, VX520, VX675 - ECR INTERFACE
VERSION 3.9.3

Verifone VX - ECR Interface

Statistics and Helper Package Communication and Message Protocol Specifications

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1. SCOPE

The scope of this document is to describe the data exchange between electronic cash-register systems (ECR) and EFT-POS terminals. The content and format of the data messages, the control and error signaling are described below.

2. OVERVIEW

The environment for the described protocol is made from the EFT-POS - ECR group devices. The basic applications used in this environment are:

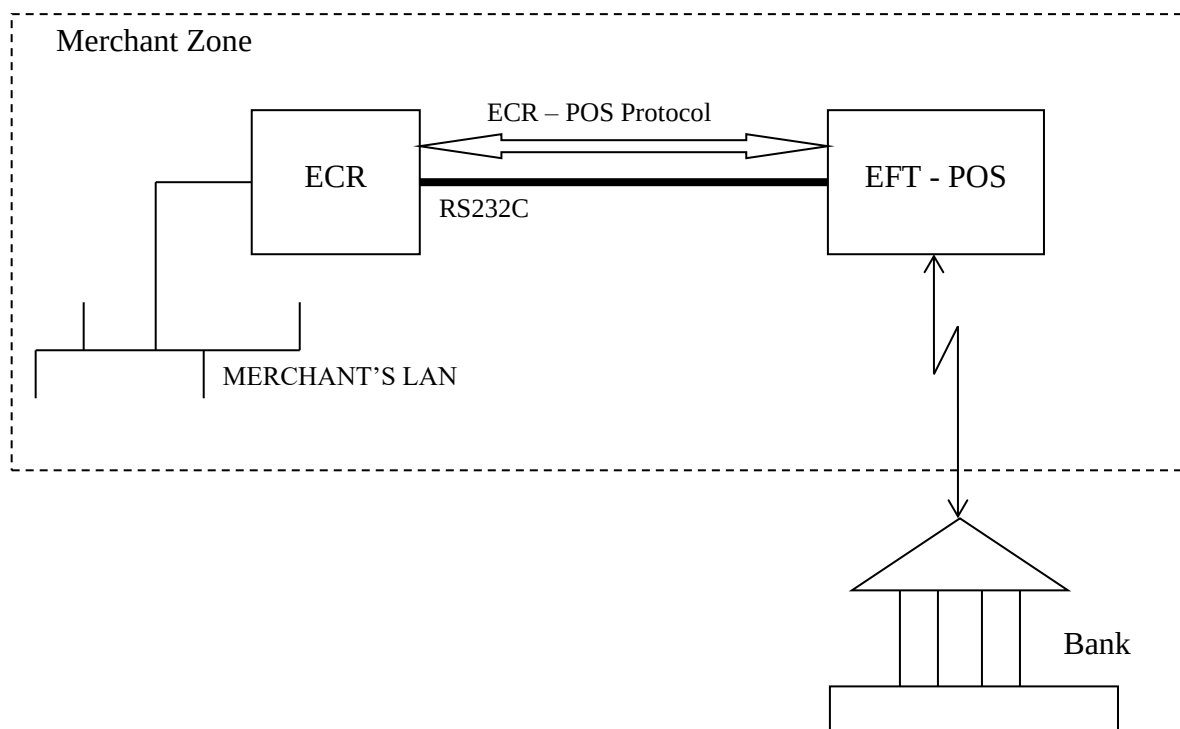
- applications that manage goods and services acquisition from the merchant;
- applications that manage fee payments to the local authorities;
- any other application which contains an ECR type system and requires card transactions processing for payment authorizations type and balance inquiry.

The definition of ECR consists of any hardware devices which meet specific corresponding function, that means the following equipment types:

- a native ECR equipment (standard)
- any PC or computing equipment with necessary software for performing any of these functions

Basically ECR and EFT-POS communicate by an asynchronous serial interface RS232C (see APPENDIX C for serial parameters). ECR equipment is connected into the merchant local area network and the EFT-POS terminals have a connection to the network of card processing server.

The simplified schema:



3. STRUCTURE OF THE DOCUMENT

More technical material is presented in the following chapters:

- Ch. 4 – Control Characters and Structuring of Message – presents a detailed description of low level protocol;
- Ch. 5 – Commands to Terminal
- Ch. 6 – Commands to ECR present ECR – EFT-POS high level protocol as it is used by the application. There are defined the commands for the ECR (Ch. 6) and to the EFT-POS (Ch. 5).

4. CONTROL CHARACTERS AND STRUCTURING OF MESSAGE

A data transmission starts with the use of *ENQ* control character and ends with *EOT* control character. *ACK* is sent to validate the received data and *NAK* to signal an issue (usually invalid *LRC*).

Control characters are the combinations *DLE STX* and *DLE ETX*. They are used as delimiters for the command messages. After the *DLE ETX* control character, the final byte is a longitudinal redundancy check (*LRC*), used by the receiver to check the command message integrity.

The *LRC* is calculated based only on the command message (without control characters) and a C example of the implementation can be found at Ch. 7.

Command messages are used to exchange data between the electronic cash-register systems (ECR) and EFT-POS terminals. A list of integrated command messages can be found at Ch. 5. The following format is used for each command:

DLE STX* <COMMAND MESSAGE> *DLE ETX* *LRC

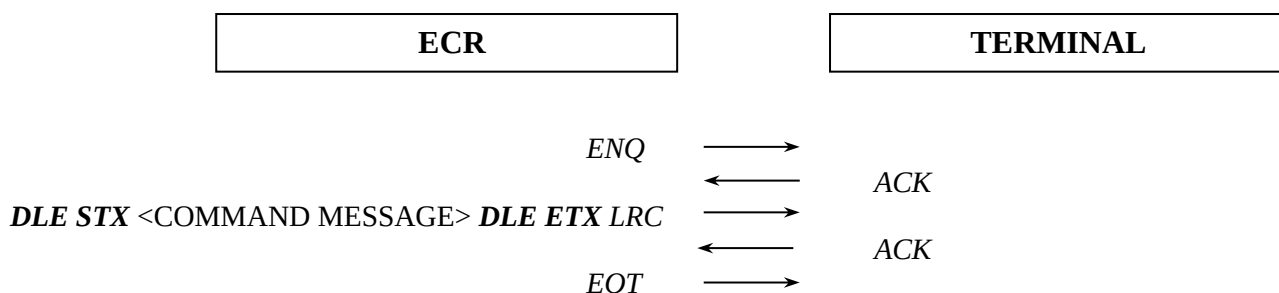
The protocol is based on two or more command messages exchanged between the electronic cash-register systems (ECR) and EFT-POS terminals. The cash register is the first to start the message exchange by the use of a specific command message, while the EFT-POS will send the rest of the commands, including the finishing one with final transaction result.

One command message exchange starts with the use of *ENQ*, is followed up by *ACK* messages and finishes with *EOT*. This is valid for both electronic cash-register systems (ECR) and EFT-POS terminals that start to send the command message.



If a *NAK* is received (either by the electronic cash-register systems (ECR) or the EFT-POS terminal), the message exchange was unsuccessful and the command message will have to be resent. After 3 consecutive *NAK*'s, *EOT* should be sent to finish the processing on both ends. In this case, the message exchange was unsuccessful and should be started from the beginning.

The first message exchange - started from the electronic cash-register:

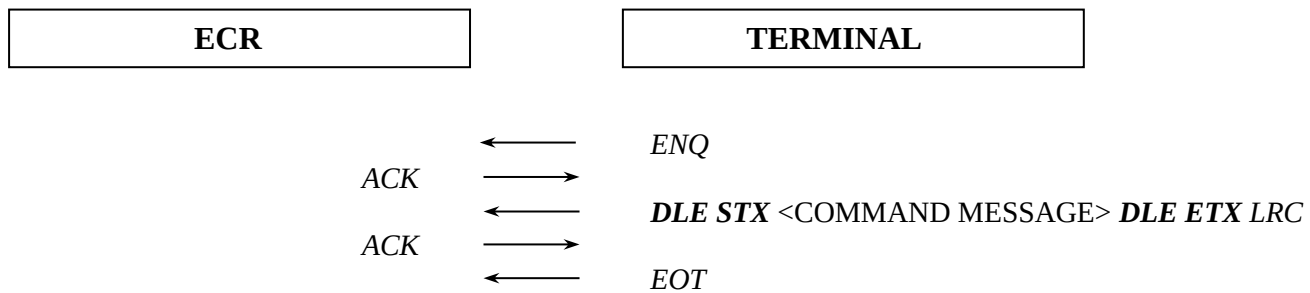


1. The EFT-POS terminal waits for an *ENQ* control character from the electronic cash-register system.
2. After receiving the *ENQ*, the EFT-POS terminal sends an *ACK* control character.
 - a. If the EFT-POS terminal does not send an *ACK* within **0.6 seconds**, the electronic cash-register system should resend the *ENQ* control character.
3. The EFT-POS terminal waits for the command message for **0.2 seconds**.
 - a. If the EFT-POS terminal does not receive the command message within this **0.2 second** interval, a timeout will be reached and the process will be considered as finished with a timeout error.
4. Upon receiving the command message, the EFT-POS terminal will check the *LRC* and send an *ACK* (the calculated *LRC* matches the received *LRC*).
 - a. If the calculated *LRC* does not match the received *LRC*, the EFT-POS terminal will send a *NAK* and wait for another command message for **0.2 seconds**.
 - b. After 3 consecutive *NAK*'s, *EOT* should be sent by the electronic cash-register system to finish the processing on both ends.
5. The EFT-POS terminal will wait for the *EOT* for **0.2 seconds**.

The recommended minimum timeout for electronic cash-register system to wait for an *ACK* reply from the EFT-POS terminal is **0.6 seconds**. Any additional latency caused by any cash-register peripherals/cables should be taken into account when considering the minimum timeout.

In case the EFT-POS terminal has reached a timeout, the protocol was not successfully finished and the message exchange has ended. The EFT-POS terminal will wait for the *ENQ* to start the first message exchange.

The second (and all the following) message exchange - started from the EFT-POS terminal:



1. The electronic cash-register has send the *EOT*, marking a successful end to the first message exchange and should **immediately** wait for an *ENQ* control character from the EFT-POS terminal. The maximum receive timeout should be set at **2 minutes** (this covers any of the possible message exchanges, including user interaction, maximum contactless card reading timeout, maximum PIN entry timeout, communication connect/send/receive timeouts set by the bank and receipt printing time).
2. After receiving the *ENQ*, the electronic cash-register sends an *ACK* control character.
 - a. If the electronic cash-register does not send an *ACK* within **0.2 seconds**, the EFT-POS terminal will resend the *ENQ* and wait for another *ACK*.
 - b. After 3 consecutive timeouts and no *ACK* received, *EOT* is sent by the EFT-POS terminal to finish the processing on both ends.
3. The electronic cash-register system waits for the command message for **0.6 seconds**.
 - a. EFT-POS terminal will send the command message after *ACK* is received.
4. Upon receiving the command message, the electronic cash-register system will check the *LRC* and send an *ACK* (the calculated *LRC* matches the received *LRC*).
 - a. If the calculated *LRC* does not match the received *LRC*, the electronic cash-register system will send a *NAK* and wait for another command message for **0.6 seconds**.
 - b. After 3 consecutive *NAK*'s, *EOT* should be sent by the EFT-POS terminal to finish the processing on both ends.
5. The electronic cash-register system will wait for the *EOT* for **0.6 seconds**.

The recommended timeout for electronic cash-register system to wait for the next message exchange from the EFT-POS terminal is **2 minutes**.

4.1 Application Protocol

The protocol differentiates between enquiries and responses. Depending on the context, one of the devices sends commands, whereas the other returns responses. It is not permitted to send commands whilst other commands are processed.

A basic element of the protocol is APDP (Application Protocol Data Package) which consists of System Field (SFLD), Data Length field (DLNG), and Data Block:

SFLD	DLNG=N	DATA
(len=2bytes)	(len=1byte)	(len=Nbytes)

4.2 System Field of Application Protocol

This contains two fields, one byte each, and each administering different operations depending on whether a command or a response is to be sent.

SFLD

Command:

0	KLASSE	INSTR
----------	---------------	--------------

Command response:

1	KKRW	APRW
----------	-------------	-------------

Where a command is to be sent fields KLASSE (class) and INSTR (instruction) are used, which together define the application command. KLASSE defines the class of the command, whereas INSTR defines the command within in the class.

The KLASSE's Most Significant Bit (MSB) is always set to zero. It indicates that this is a command.

Where a response is to be returned the fields used are KKRW (Class Command of Returned Value) and APRW (Application Program Returned Value). MSB of KKRW is always set to one. KKRW indicates the total positive or negative result from executing the command, whereas APRW indicates the status of the returned result.

A positive result can be defined in two ways. In the first case KKRW=80h which means a successful result. In the second case, where KKRW=84h, APRW field has to be tested and if it contains 00h this also indicates a positive result.

For KKRW field 80h and 84h values are used only.

4.3 Data Length Field of Application Protocol

This field contains the length of the data block which follows it. In this protocol the only field length to be used is 1 byte and the permitted values are from 0 to 254. Length 255 should not be used since with the original protocol this will mean extension of the field to 2 bytes. This extension is not yet developed.

4.4 Data Block of Application Message

This contains the application information. Its length is defined by DLNG field.

4.5 Classes of Commands

Currently only EXECUTE class is supported with code 06.

4.6 Coding of Responses

Responses are coded in two variants.

Where KKRW=80h this indicates a positive response to the command which has triggered it. In this case the field APRW is to be disregarded and is usually =00.

Where the response is negative KKRW=84h meaning that APRW field is valid. In this case APRW contains error code.

KKRW=84h and APRW=00h are peculiar cases indicating positive response.

Table of Error Codes in APRW Field:

Error Code	MEANING
0	Received command , processing started
2	ECR has not executed login
3	Unknown command in INSTR field

5. COMMANDS TO TERMINAL

These commands are sent from the ECR to the terminal.

Command	Control Field		Data Fields						
	Class (hex)	Instr.-ns. (hex)	Amount	Additional Amount	Currency	Article code	Quantity	Receipt Number	RRN
Login	06	0x00	-	-	-	-	-	-	-
Logout	06	0x02	-	-	-	-	-	-	-
Sale	06	0x01	✓	-	-	✓	✓	-	-
Sale with Cashback	06	0x08	✓	✓	-	-	-	-	-
Verify	06	0x03	-	-	-	-	-	✓	-
Void	06	0x30	✓	-	-	-	-	✓	-

5.1 Data Fields Description

<i>Name</i>	<i>Length (bytes)</i>	<i>Obs.</i>
[Amount]	12AN	with leading zeroes
[Additional amount]	12AN	with leading zeroes
[Currency]	3A	currency name
[Article code]	3AN	if is not used set to '000'
[Quantity]	6AN	with leading zeroes if is not used set to '000000'
[Receipt Number]	6AN	with leading zeroes
[RRN]	12AN	with leading zeroes if is not used set to '000000000000'

5.2 Commands

5.2.1 Login

The terminal can process the rest of the below commands only after having received this message. You can operate the terminal provided after powering it on the message is sent at least once.

ECR -> terminal

KLASSE = 06
INST = 00
DLNG = 00

terminal -> ECR

KKRW = 80/84
APRW = XX
DLNG = 00

The electronic cash-register will send this command within the first message exchange (see Ch. 4). A successful message exchange of the Login command will trigger the EFT-POS to send the response

back (KKRW = 80) to the electronic cash-register. Any message exchange error will trigger the EFT-POS to send the error response back (KKRW = 84) to the electronic cash-register.

The Login command should be sent (by the ECR to terminal) **at least once**: either after the terminal boots up, either after a Logout command was previously sent by the cash-register to the terminal.

5.2.2 Logout

After performing this command the terminal can only receive Login command or another logout request.

ECR-> terminal	terminal->ECR
KLASSE = 06	KKRW= 80/84
INST = 02	APRW= XX
DLNG = 00	DLNG = 00

The electronic cash-register will send this command within the first message exchange (see Ch. 4). A successful message exchange of the Logout command will trigger the EFT-POS to send the response back (KKRW = 80) to the electronic cash-register. Any message exchange error will trigger the EFT-POS to send the error response back (KKRW = 84) to the electronic cash-register.

5.2.3 Sale

The ECR places a request for a sale transaction and must transmit the amount.

ECR -> terminal	terminal -> ECR
KLASSE = 06	KKRW = 80/84
INST = 01	APRW = XX
DLNG = LL	DLNG = XX
Data ...	

The electronic cash-register will send this command within the first message exchange (see Ch. 4). A successful message exchange of the Sale command will trigger the EFT-POS to send the response back (KKRW = 80) to the electronic cash-register. Any message exchange error will trigger the EFT-POS to send the error response back (KKRW = 84) to the electronic cash-register.

The sale command does not currently use the content of article code and quantity fields and the electronic cash-register must send the default values ("000" for article code and "000000" for quantity). Please see *Appendix A* for a practical example.

After a successful first message exchange, the EFT-POS will continue with the second (and all the following) message exchange, described below:

1. The first step is to read the card data, with a maximum timeout of **2 minutes**.

2. The second step is PIN entry, with a maximum timeout of **1 minutes**. The EFT-POS will send a message to the ECR to inform that PIN is about to be entered. PIN entry is an optional step, in some cases the PIN is not required to be entered and in these cases the PIN entry message is not sent to the ECR.
3. The third step is the communication with the bank, with a maximum timeout of **2 minutes**. The EFT-POS will send a message to the ECR to inform that communication with the bank will start. Transactions can also be offline approved (without communication with the bank), meaning this step can also be skipped.
4. The EFT-POS will send the final command (indicating either transaction end/approval, either refusal) to the ECR, along with additional information (for more information please see Ch. 6).

**Please keep in mind that the bank can make changes to the overall timeouts, including communication timeouts of the EFT-POS.*

5.2.4 Void

The ECR places a request for void an original sale. ECR must transmit the amount for original sale and the receipt number of a previous transaction.

ECR -> terminal	terminal -> ECR
KLASSE = 06	KKRW = 80/84
INST = 0x30	APRW = XX
DLNG = LL	DLNG = XX
Data ...	

The electronic cash-register will send this command within the first message exchange (see Ch. 4). A successful message exchange of the Void command will trigger the EFT-POS to send the response back (KKRW = 80) to the electronic cash-register. Any message exchange error will trigger the EFT-POS to send the error response back (KKRW = 84) to the electronic cash-register.

The Void command contains the amount and the receipt number of original/previous sale transaction. Please see *Appendix A* for a practical example.

After a successful first message exchange, the EFT-POS will continue with the second (and all the following) message exchange, described below:

1. The first step in the terminal application is to ask the customer/merchant to confirm the amount of the original transaction using the EFT-POS interface.
2. The second step is to read the card data, with a maximum timeout of **2 minutes**.
3. The third step is the communication with the bank, with a maximum timeout of **2 minutes**.
4. The EFT-POS will send the final command (indicating either transaction end/approval, either refusal) to the ECR, along with additional information (for more information please see Ch. 6).

**Please keep in mind that the bank can make changes to the overall timeouts, including communication timeouts of the EFT-POS.*

6. COMMANDS TO ECR

These commands are sent from the terminal to the ECR.

Command	Control Field		Data Fields
	Class (hex)	Instr-ns. (hex)	
Authorization End	06	0F	see APPENDIX B
Refusal	06	1E	-
Pin Entry	05	01	-
BeginAuth	05	02	-

6.1 End of Processed Request

This is a positive command for completing data processing requested by the ECR.

terminal->ECR	ECR->terminal
KLASSE = 06	KKRW = 80/84
INST = 0F	APRW = XX
DLNG = LL	DLNG = 00
Data ...	

This command logically ends the procedure started with a command from the ECR.

6.2 Refusal

With this command the terminal completes processing the ECR's request which has not been executed due to an error. The error is indicated in the 'error code' field.

terminal->ECR	ECR->terminal
LASSE = 06	KKRW = 80/84
INST = 1E	APRW = XX
DLNG = 01	DLNG = 00
Error code ...	

6.3 PIN Entry

This is an information command to the ECR. The POS send this info before the PIN is entered.

terminal->ECR
KLASSE = 05
INST = 01
DLNG = 00

6.4 Begin Auth

This command informs the ECR that POS begin communication with the bank.

terminal->ECR

KLASSE	= 05
INST	= 02
DLNG	= 00

7. CALCULATION OF THE CHECKSUM

LRC Calculation - 'C' style

```
static byte calcLRC( byte *buf, int len )
{
    byte b=0; int i;

    for(i=0; i<len; b = b ^ buf[i], i++);
    return b;
}
```

8. FINAL CONSIDERATION

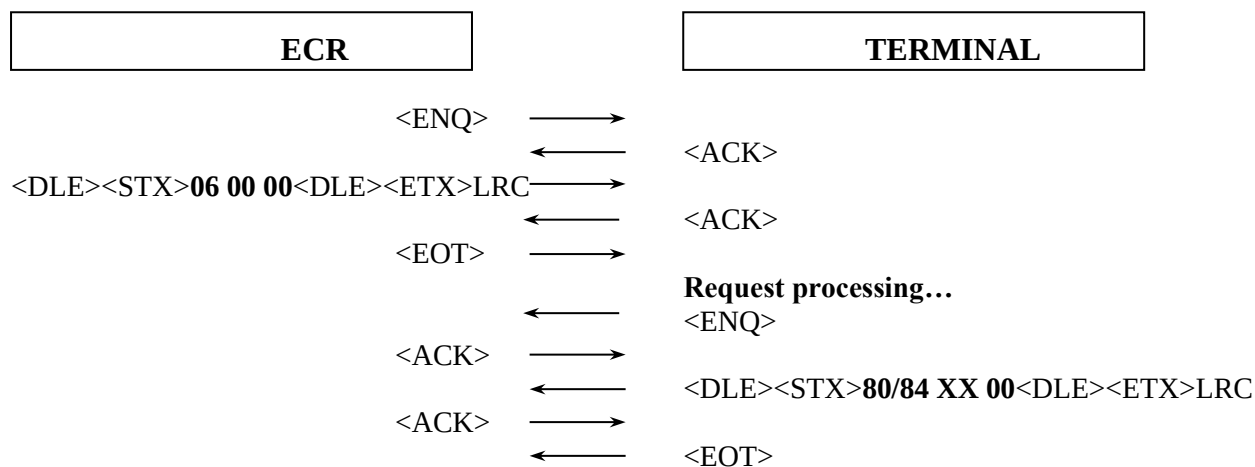
This material contains the technical documentation for the version V3.9.2 of ECR – EFT-POS protocol.

The future versions of the protocol can implement more functionalities and commands, conserving the messages formatting compatibilities exchanged between ECR and EFT-POS with previous versions. Any future version will have a revised documentation.

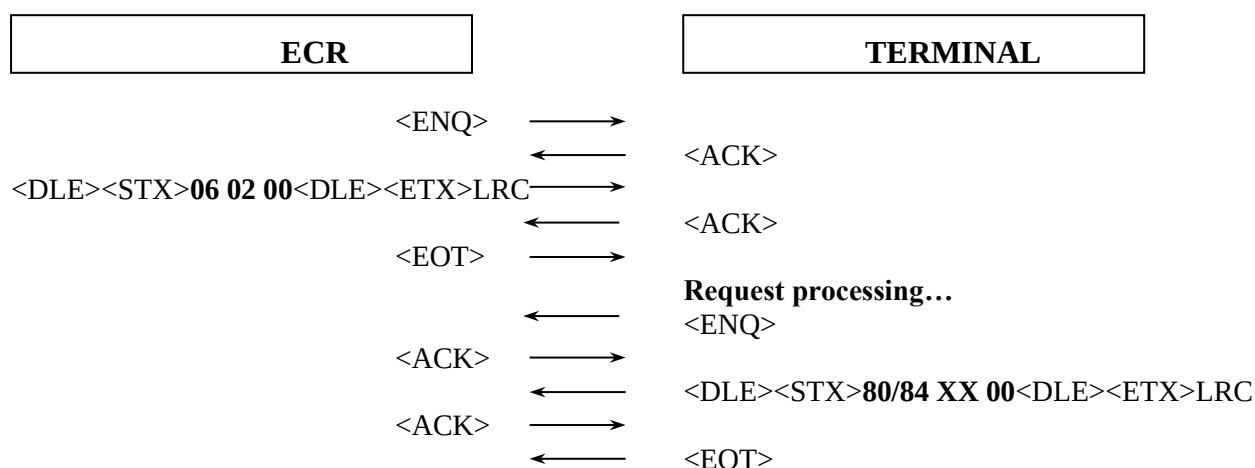
In the phase of new application development for an ECR – EFT-POS system, knowing the big differences between the software capabilities and flexibilities of the two components, it is recommended that only ECR software application to be subject for later adjustments according to the existing protocol and client requests.

APPENDIX A

LOGIN

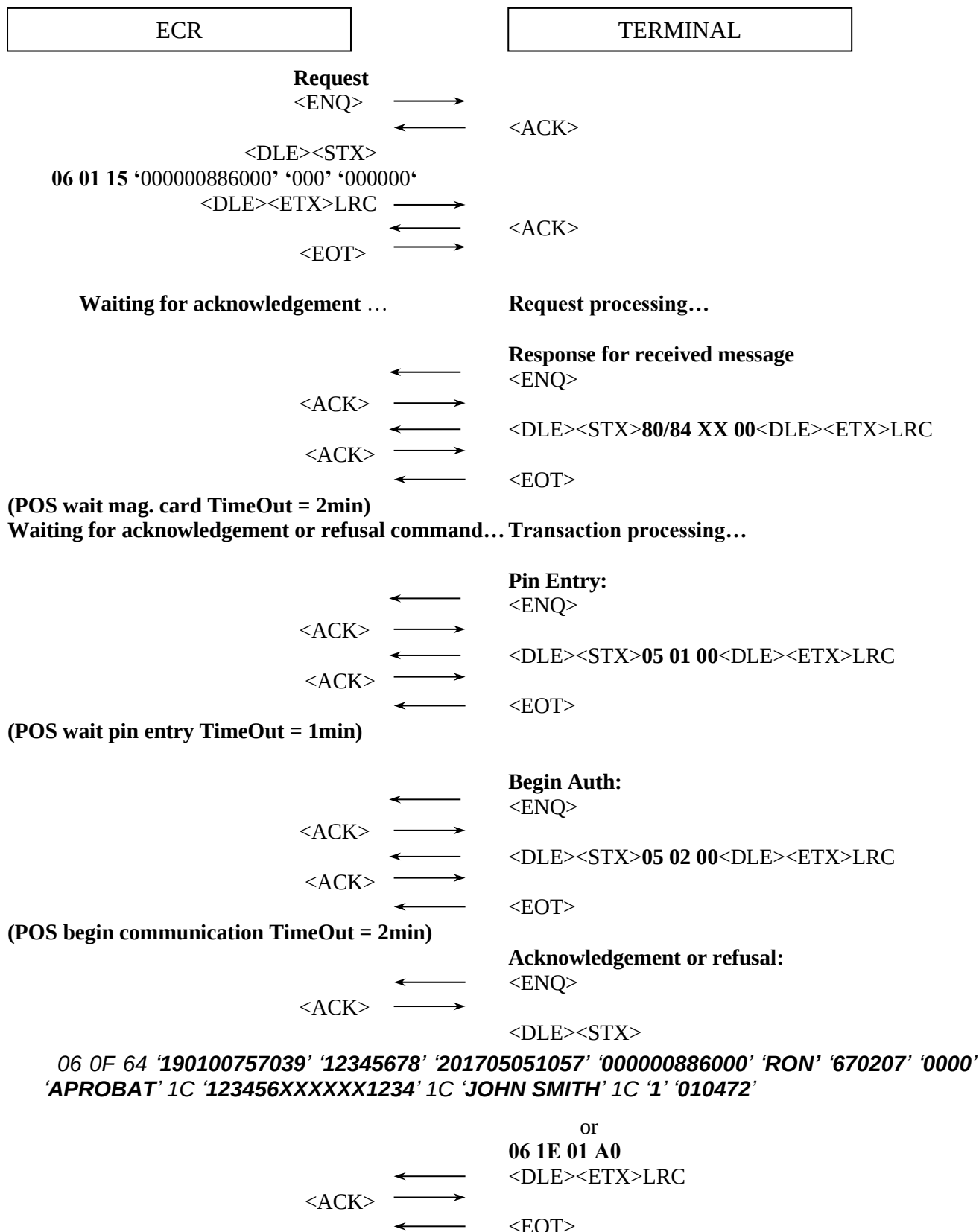


LOGOUT



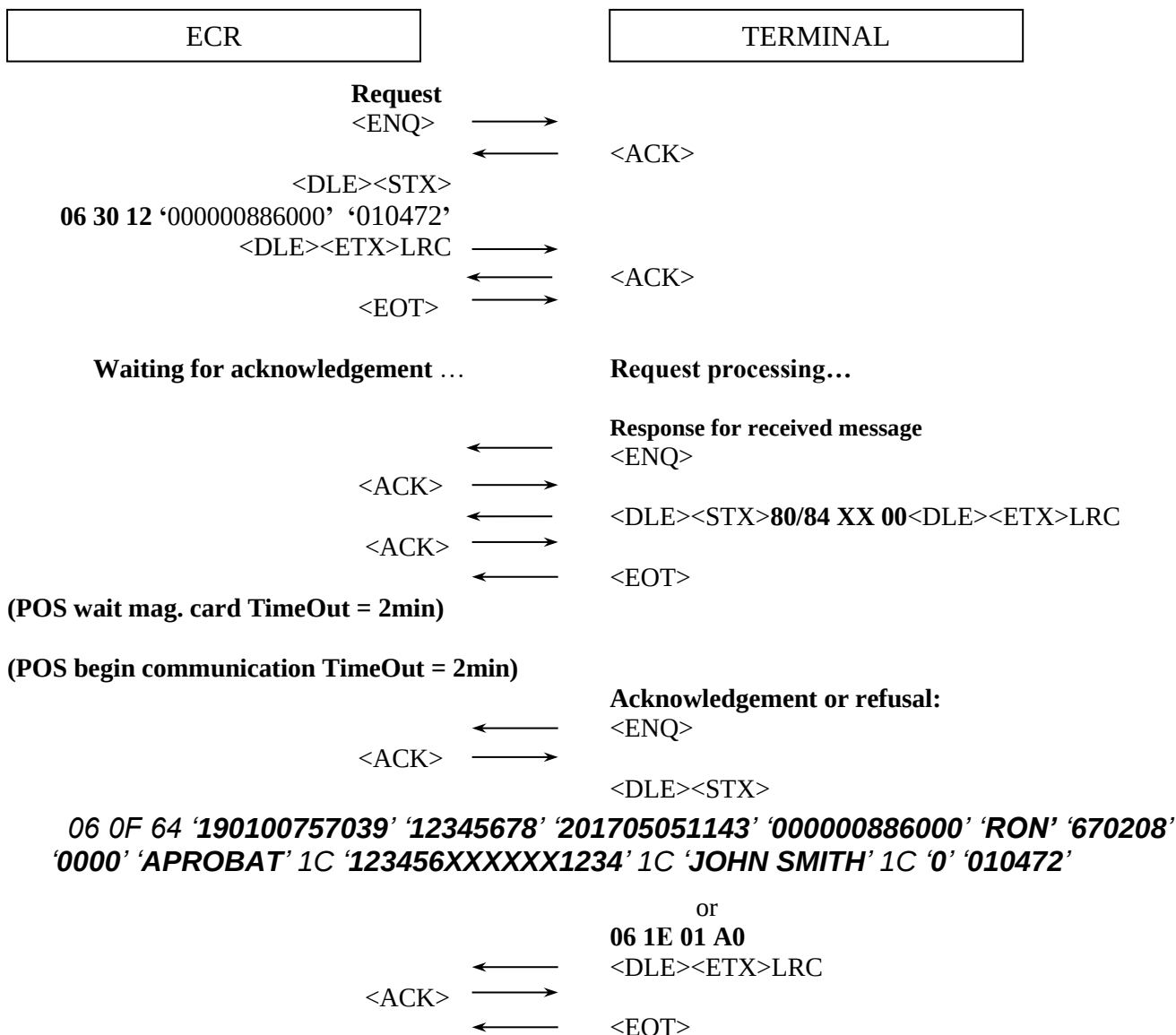
SALE

Article code **000**, Quantity **000000**, Amount **8860.00 RON**



VOID

Receipt Number **010472**, Amount **8860.00 RON**



APPENDIX B

1. Login and Logout functions remain unchanged
2. **Sale, Sale with Cashback, Verify, Void:**

After transaction processing the packet POS ->ECR will have the following structure:

```
<DLE><STX>
06 0F XX [Reference number] [Terminal id] [Transaction Date] [Amount] [Currency]
[Authorization Code] [Response Code] [Response Code Text] [Card No.] [Card Holder Name] [Pin
Identifier] [Receipt No.]
<DLE><STX><LRC>
```

Description:

<i>Name</i>	<i>Length (bytes)</i>	<i>Obs.</i>
XX	1H	
[Reference number]	12AN	with leading zeroes
[Terminal ID]	8AN	with leading zeroes
[Transaction Date]	12AN	12AN format: YYMMDDhhmmss
[Amount]	12AN	with leading zeroes
[Additional Amount]	12AN	with leading zeroes
[Currency]	3A	currency name
[Authorization Code]	6AN	with leading zeroes
[Response Code]	4AN	with leading zeroes
[Response Code Text]	Max 60A	Terminator 0x1C
[Card No.]	Max 30A	Terminator 0x1C
[Card Holder Name]	Max 40A	Terminator 0x1C
[Pin Identifier]	1AN	1=with pin;0=without pin
[Receipt No.]	6AN	with leading zeroes

Characters notation:

- A ASCII
- AN ASCII Numeric
- H Hex

Note:

1. XX = data length between [Reference number] and last field in packet.
2. LRC is calculated for data with XX length.
3. If a fixed length field is empty data from the field will be filled with 0x30.

[Reference number] is filled with 0x30 when the transaction is offline.

[Response Code Text] and [Card Holder Name] are not always present.

A transaction online is approved when:

- i. [Response Code] is 0000
- ii. [Authorization Code] has been received with 6 alphabetic and numeric characters (the letters A – Z and numbers 0 - 9)
- iii. [Reference number] has been received with 12 alphabetic and numeric characters (the letters A – Z and numbers 0 - 9)

** The reference number is assigned by the host to uniquely identify a transaction. It has an alphanumeric format*

*** The authorization code is the unique (within a business day) response identification value assigned by the authorizing institution*

APPENDIX C

Serial communications parameters.

Speed	9600 bps
Data bits	8
Parity	None
Stop bits	1
Flow control	None

APPENDIX D

Binary	Oct	Dec	Hex	Abbr
000 0000	000	0	00	NUL
000 0001	001	1	01	SOH
000 0010	002	2	02	STX
000 0011	003	3	03	ETX
000 0100	004	4	04	EOT
000 0101	005	5	05	ENQ
000 0110	006	6	06	ACK
000 0111	007	7	07	BEL
000 1000	010	8	08	BS
000 1001	011	9	09	HT
000 1010	012	10	0A	LF
000 1011	013	11	0B	VT
000 1100	014	12	0C	FF
000 1101	015	13	0D	CR
000 1110	016	14	0E	SO
000 1111	017	15	0F	SI
001 0000	020	16	10	DLE
001 0001	021	17	11	DC1
001 0010	022	18	12	DC2
001 0011	023	19	13	DC3
001 0100	024	20	14	DC4
001 0101	025	21	15	NAK
001 0110	026	22	16	SYN
001 0111	027	23	17	ETB
001 1000	030	24	18	CAN
001 1001	031	25	19	EM
001 1010	032	26	1A	SUB
001 1011	033	27	1B	ESC
001 1100	034	28	1C	FS
001 1101	035	29	1D	GS
001 1110	036	30	1E	RS
001 1111	037	31	1F	US
111 1111	177	127	7F	DEL